Manuscript outline

Cell enlargement, polarity competition and migration persistence

Polarity replacement, spatial response maps and noise-driven heading memory

23 October 2026 · Size, polarity susceptibility and heading persistence

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations support this hypothesis; new paired WGD motility results retain a substantial, explicitly reserved contribution.

Figure 2 separates size-dependent initiation and cue-width responses from temporal class signatures: initiation-limited resetting, saturating activation with visible gap retention, and pulse-2 progress after loss of visible activity. The full Holmes width series shows finite replacement at all tested lengths for sufficiently broad cues; narrow-cue censoring cannot establish a growing penalty.

Enlargement slows both cue-driven turning routes and selectively suppresses edge susceptibility, most strongly in Wave-pinning and Goryachev. A native spatial response map, tested away from the edges, localizes nearly all predicted drift to the two lateral boundaries. Without fitting noisy drift, the calculation predicts Wave-pinning’s C60/C20 drift ratio of 0.490 versus 0.473 measured and a persistence size exponent above 2; Goryachev’s absolute drift prediction fails quantitatively.

Enlargement extends model heading memory and raises noise tolerance, while the size advantage weakens as input strengthens. Shared translation explains eligible weak-input Wave-pinning, Holmes and Spring conditions; all-input closure bounds, remainder curves and fixed-follow-up takeover incidence show where the domain description becomes incomplete. Fourier-mode agreement is partial and sensitive to local-slope estimation; an explanation covering all four strong-input models is not established.

The prospective pilot table fixes model-linked cellular observables before the SUM-159 data arrive. Directions are conditional hypotheses; unavailable model measurements remain unavailable. Activity heading, observed migration summaries and conditional OU

Manuscript outline · continued

velocity-memory estimates remain separate evidence.

This is a working manuscript outline. Figures 1–2 retain their numerical evidence with separate panels for distinct quantities. The published migration evidence in Figure 5 and reserved Figure 6 retain their scientific content. The prospective pilot directions retain their operational definitions with updated panel references and model-conditioning notes. Activity polarity, modelled replacement, observed migration and conditional OU velocity memory remain distinct readouts.

Proposed main figures

- Figure 1. Enlargement changes the cue duration required for lasting polarity replacement
- Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures
- Figure 3. Enlargement slows both turning routes and selectively suppresses edge susceptibility
- Figure 4. Enlargement extends polarity heading memory under fluctuating input
- Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation
- Figure 6. RESERVED: new WGD motility-assay results

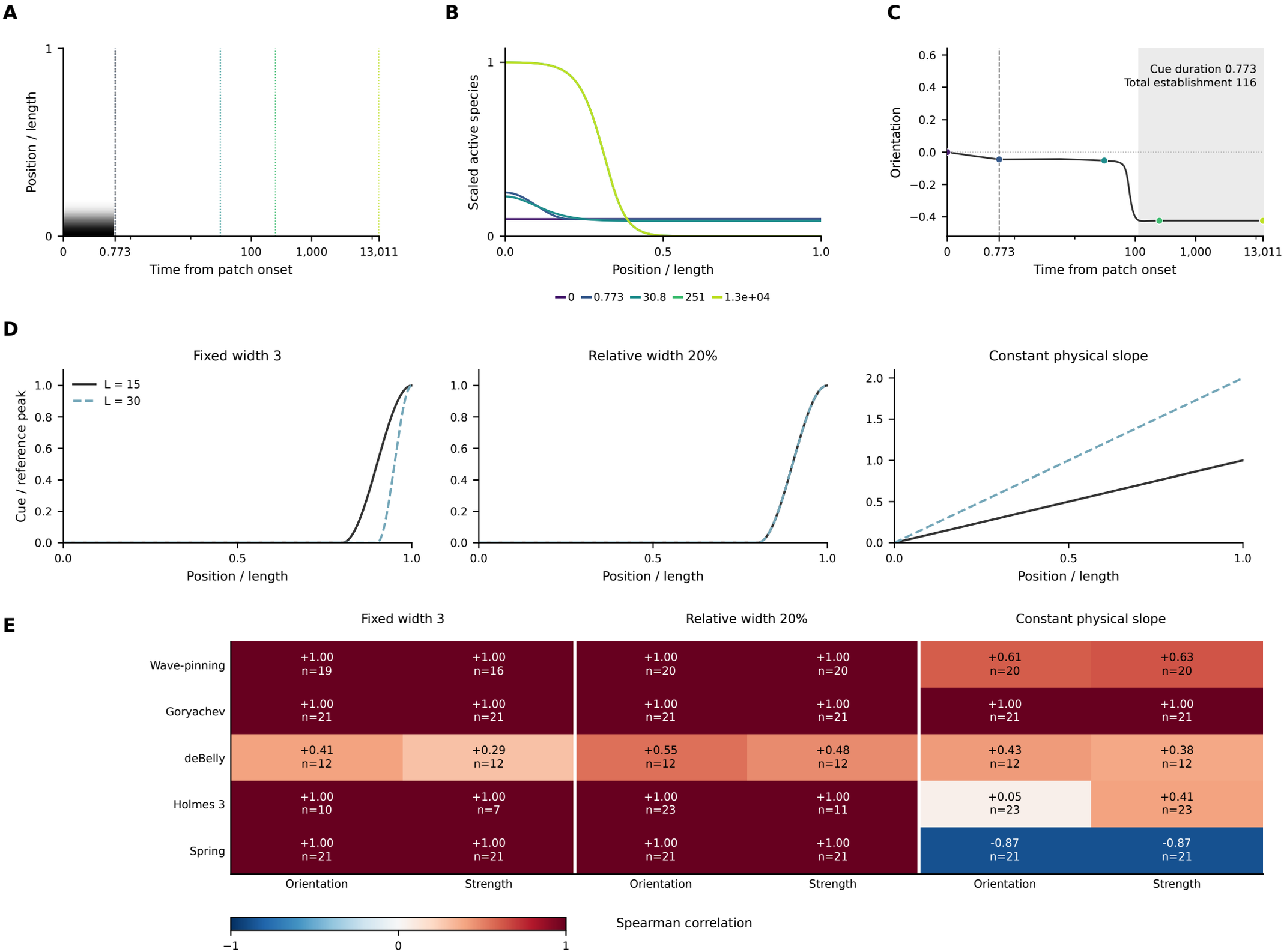

A
Position / length
Time from patch onset
0
0.773
100
1,000
13,011
B
Scaled active species
Position / length
0
0.773
30.8
251
1.3e+04
C
Orientation
Cue duration 0.773
Total establishment 116
Time from patch onset
D
Fixed width 3
Relative width 20%
Constant physical slope
Cue / reference peak
L = 15
L = 30
Position / length
E
Fixed width 3
Relative width 20%
Constant physical slope
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
+1.00 n=19
+1.00 n=16
+1.00 n=20
+1.00 n=20
+0.61 n=20
+0.63 n=20
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+0.41 n=12
+0.29 n=12
+0.55 n=12
+0.48 n=12
+0.43 n=12
+0.38 n=12
+1.00 n=10
+1.00 n=7
+1.00 n=23
+1.00 n=11
+0.05 n=23
+0.41 n=23
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
-0.87 n=21
-0.87 n=21
Orientation
Strength
Spearman correlation

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

Comparative theoretical foundation: five established models, three finite challenges, and two distinct scalar replacement gates.

Detailed legend

A, Recorded Wave-pinning preparation cue at length 15: a finite left patch, withdrawn at 0.773 native time units. White to black is zero to peak input; coloured vertical lines mark the profile samples in B.

B, Native active profiles from the same preparation, scaled using the common minimum and span of the displayed snapshots. Colours identify recorded times.

C, Orientation from the same preparation. Grey marks basal verification after establishment arrival at about 116; the cue has already ended. The matched time scale in A and C separates input duration from autonomous development.

D, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

E, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

deBelly uses independently commanded corrected endpoints, rather than the deprecated in-loop centered extension. Spring length and time retain native units and are not silently converted to micrometres or seconds. Model-specific axes and stimulus definitions prevent interpreting cross-model amplitudes or clocks as biologically calibrated equivalents.

Narrative points and conclusions

- Fixed-width and relative-width patches show perfectly increasing rank order among observed successes in Wave-pinning, Goryachev, Holmes 3 and Spring; enlargement often increases the finite stimulation required for a retained response.
- The response depends on the cue geometry and model. Spring has a negative constant-slope association (Spearman correlation -0.874 for both gates), while Holmes orientation under the constant-slope cue is nearly unassociated with length (0.047).
- Corrected deBelly curves are nonmonotone despite moderate positive rank correlations. The full duration curves and eligibility ranges are essential to the result.
- These are deterministic polarity-replacement experiments. They motivate, but do not measure, spontaneous polarity memory or migration persistence.

Proposed measurements and experimental follow-up

Mechanism-linked cellular follow-up would match local input support, amplitude and duration, track the original and challenged activity regions through stimulation and withdrawal, and distinguish orientation replacement from recovered activity strength. This validation is proposed and has no experimental results in this outline.

Scope, limitations and pending validation

Correlations are conditional on observed scalar success, rather than an unbiased summary of all possible sizes or a universal scaling law. Ineligible starts are distinct from responses unobserved by the finite horizon.

A retained scalar response can coexist with a changing native field. Only 13 of the 72 corrected deBelly scalar successes pass the separate native-field tail flag. Resolution checks from the earlier centered-deBelly campaign cannot qualify the corrected replacement thresholds.

Sources and reproducibility

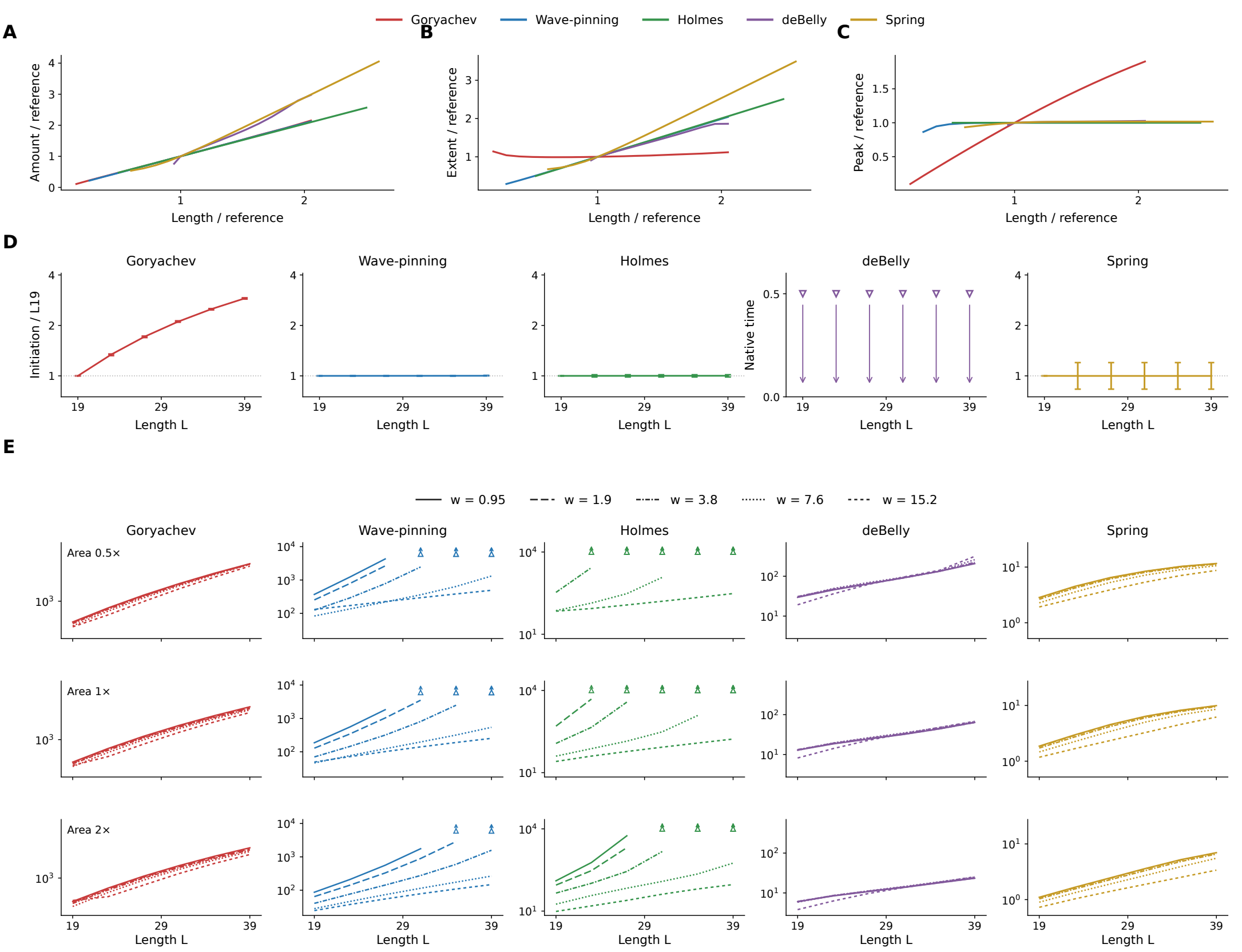
Three-test protocol and Spearman rule

Correlation table

Corrected endpoint method

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

Current duration curves


Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
B
Extent / reference
Length / reference
C
Peak / reference
Length / reference
D
Goryachev
Wave-pinning
Holmes
deBelly
Spring
Initiation / L19
Native time
Length L
E
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
Area 0.5×
Area 1×
Area 2×
Replacement time (native units)
Length L
F
Pulse-2 gain / headroom
Gap-end / pulse 1
Open circles: below activity gate at x = 0. Size increases from light (L19) to dark (L39).

G
Goryachev
lost
Wave-pinning
kept
Holmes
kept
deBelly
lost then commits
Spring
lost then commits
Challenger share
Time / pulse
L39
H
WGD−
WGD+
Initiation-limited
Goryachev
Saturating activation
Wave-pinning / Holmes
Slow non-activity state
deBelly / Spring
Biosensor
(schematic)
Time (schematic units)
I
Replacement time
(schematic)
narrow
broad
Width at fixed dose

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures

Separate size-dependent initiation and cue-width responses from observable pulse-memory classes.

Detailed legend

Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.

A, Incumbent excess amount normalized to the model reference, across the same 97 retained states.

B, Incumbent extent normalized to the model reference, for those same states.

C, Incumbent peak normalized to the model reference, for those same states.

D, Initiation is the time from cue onset until maximum in-cue activity minus the profile minimum exceeds 10% of the model's L19 incumbent excess peak. Area 1, width 3.8; times are normalized to L19. All six deBelly upper bounds are 0.5 native units (downward arrows); first-update sampling leaves its size dependence unresolved.

E, Replacement time (first eligible challenger share ≥ 0.75) versus length, with a line per width and rows for areas 0.5, 1 and 2. Upward triangles/arrows are lower bounds: the cue never replaced within the horizon (6,000 Wave-pinning; 10,000 Goryachev/Holmes; 2,000 deBelly; 100 Spring, native units). Broad cues rescue Holmes across all lengths; narrow-cue censoring does not establish a growing penalty. Spring native widths equal the displayed widths divided by 38.

F, Gap-1 activity kept versus pulse-2 gain as a fraction of headroom (Eq. 1). Four equal pulses, total ON 1.5 times the model-length commitment calibration; gaps 0–16 pulse durations. Holmes L31–39 uses the L27 calibration. Other models are grey; lightness indicates L. Open circles at x = 0 code activity below the existing visibility gate. At the left wall, deBelly/Spring gain on pulse 2 despite no visible activity; Goryachev never does. The deBelly L19/gap16 first-gap endpoint was checked against native fields.

G, L39 gap4 challenger-share histories, labelled kept / lost / lost then commits; shading marks cue ON. deBelly includes full recovery; others show one post-train pulse duration. Continuous raw shares include below-gate values. deBelly's later timing and fields remain numerically sensitive (supplement).

H, Prospective biosensor histories for the same pulse train in WGD− and enlarged WGD+ cells. Initiation-limited resetting (Goryachev), saturating activation with gap retention (Wave-pinning/Holmes), and progress retained after activity loss (deBelly/Spring) give distinct dimensionless schematic histories.

I, Prospective equal-dose width responses for those classes: weak width rescue, strong width rescue, and weaker width rescue. Solid grey: WGD−; dashed black: WGD+. These

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures · continued

are qualitative assay predictions.

Narrative points and conclusions

- Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.
- The full width series shows that Holmes can replace at every tested length when the cue is broad enough; narrow-cue censoring cannot establish a growing width-penalty ratio.
- Pulse-2 headroom gain separates low visible retention with residual progress from the initiation-limited reset response. Saturating activation retains visible activity through gaps.

Proposed measurements and experimental follow-up

Measure first in-cue biosensor rise, then vary width at fixed integrated dose.

Measure pulse-1, gap-1 and pulse-2 challenger shares under the same pulse train, followed by eventual commitment.

Scope, limitations and pending validation

Commanded input areas and native clocks are not calibrated across models.

deBelly initiation is unresolved within the first native update; later-gap timing and endpoint fields remain numerically sensitive.

F codes known below-gate activity at x=0; the exact raw ratio remains in the plotted-value CSV.

Holmes large-length trains have no target-length commitment calibration; visible retention differs from eventual replacement.

The three classes describe selected deterministic regimes; WGD biosensor predictions are prospective.

Model definitions and readouts

$$x = \frac{s_{\mathrm{gap}\,1}}{s_1}, \qquad y = \frac{s_2 - s_1}{1 - s_1} \tag{1}$$

s_1 and s_2 are challenger shares at pulse ends 1 and 2; s_gap1 is the first-gap-end share. An observed endpoint below the existing visibility gate is coded as zero visible share in F; its raw value remains in the CSV.

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures · continued

Sources and reproducibility

All 450 per-width replacement times

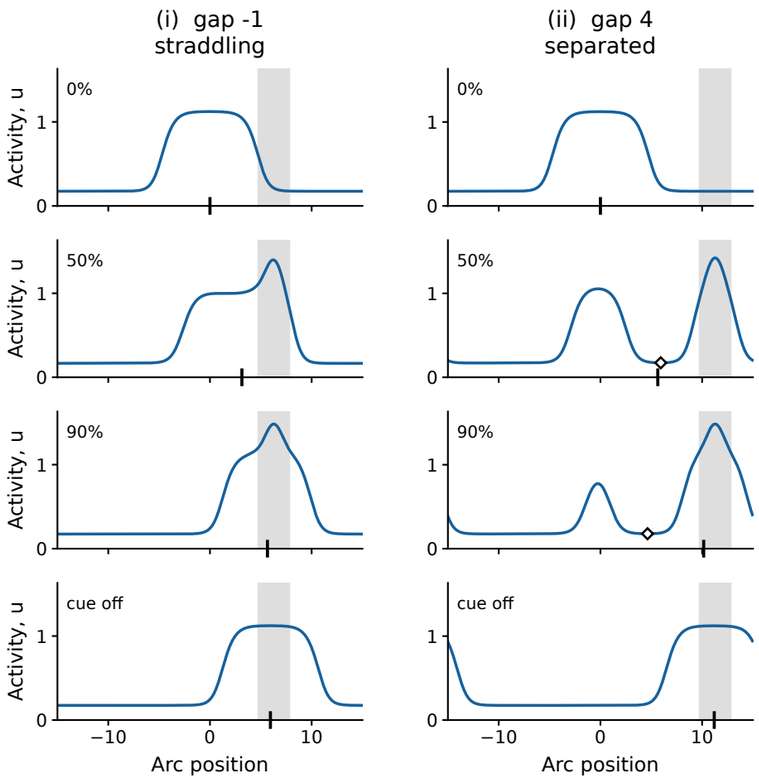
All 180 gap1/pulse2 coordinates and raw shares

deBelly L19/gap16 native-phase audit

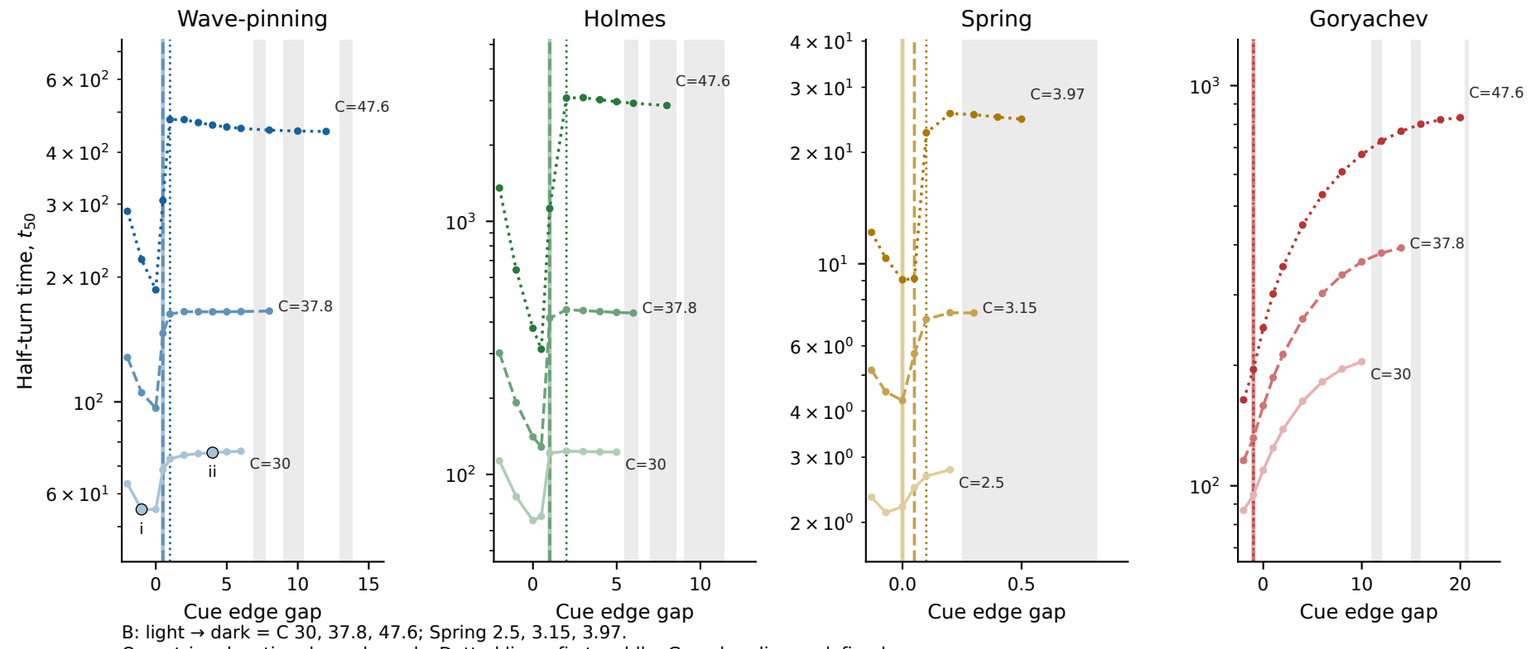
Preserved prior main panels and complete supplement

A

Grey: cue support; diamonds: saddle; | heading



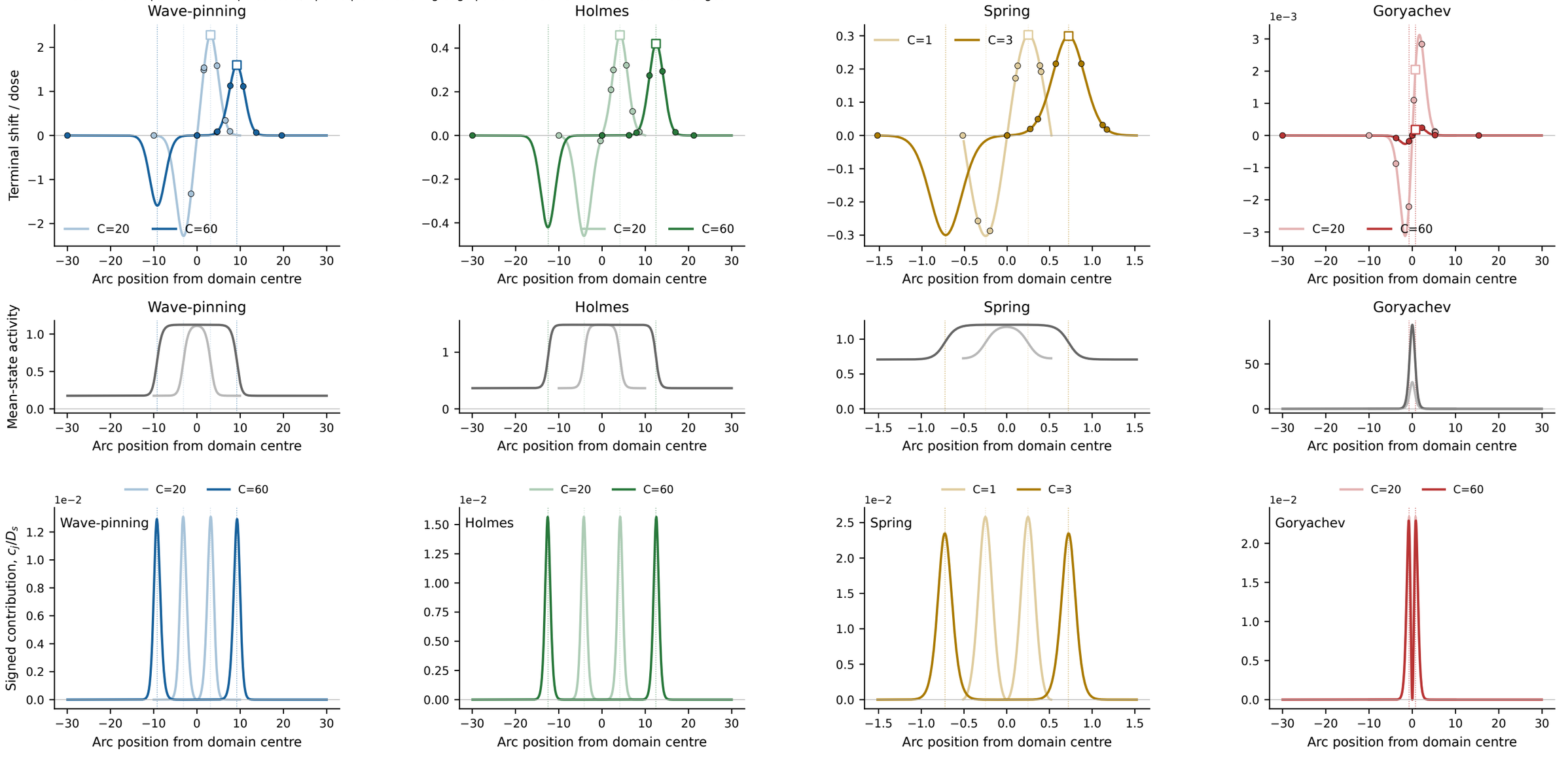
B



B: light → dark = C 30, 37.8, 47.6; Spring 2.5, 3.15, 3.97.
Open triangles: time lower bounds. Dotted lines: first saddle. Grey: heading undefined.

C

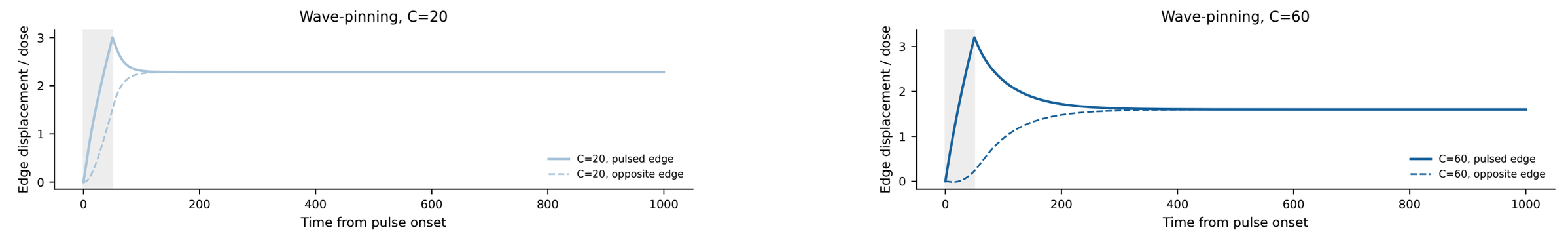
Lines: native map; circles: new pulse tests; open squares: existing edge pulse. Dotted verticals: half-contrast edges.



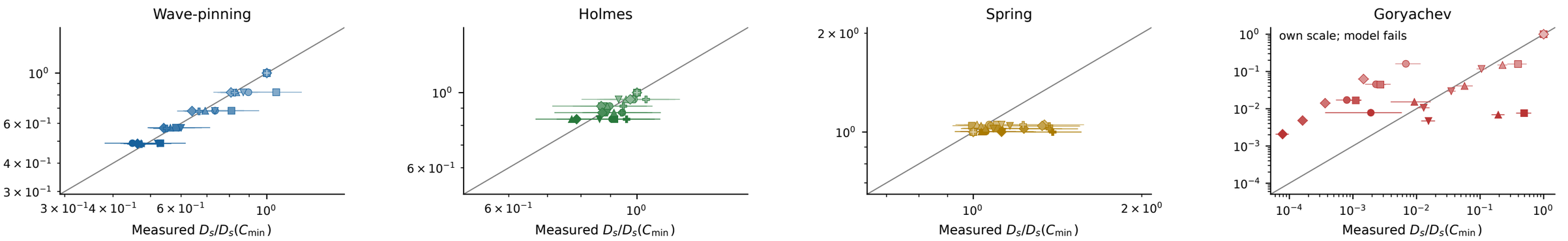
D

E

F



G



H

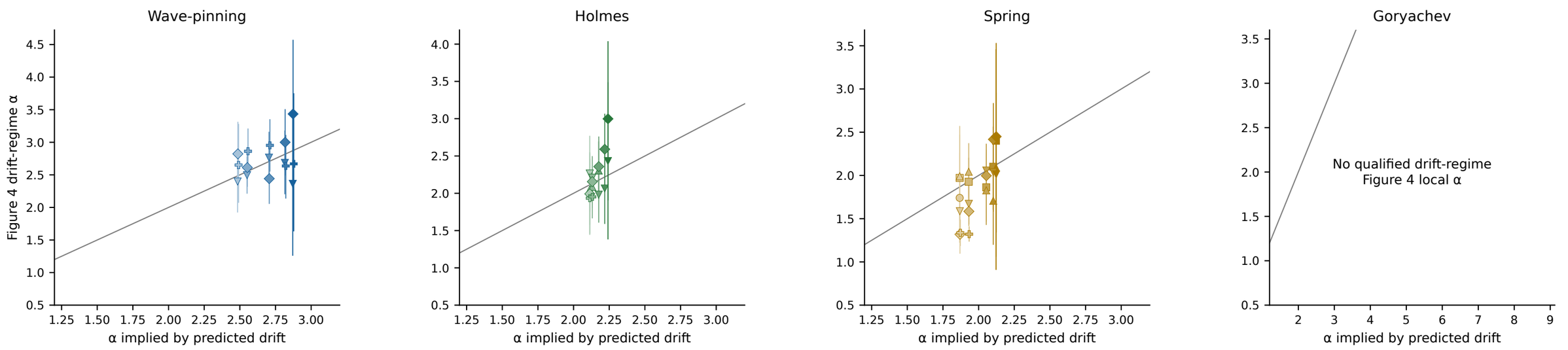


Figure 3. Enlargement slows both turning routes and selectively suppresses edge susceptibility

Size slows both turning routes; selective edge susceptibility links imposed perturbations to domain drift and excess heading persistence.

Detailed legend

A. Wave-pinning activity profiles for a cue straddling the incumbent edge (i) or separated from it (ii), C=30. Rows show the initial state, half-turn, 90% turn and cue-free relaxation. Grey marks cue support, diamonds mark saddles and the black tick marks activity heading.

B. Enlargement slows half-turning through both incumbent sliding and replacement by a new region. Half-turn time is plotted against cue-edge gap for C=30, 37.8, 47.6 (Spring 2.5, 3.15, 3.97), light to dark. Vertical lines mark the first saddle at half-turn; grey marks undefined heading at one or more sizes. Open upward triangles with arrows denote time lower bounds.

C. Signed terminal displacement per dose for a Gaussian pulse centred at each arc position, at f=0.001: SD 1.5, duration 50 (Spring 0.15, 5). Light and dark are C20/C60 (Spring 1/3). Lines are the native linearized response to this pulse shape; circles are direct pulse tests and open squares mark the edge test. The edge pulse confirms the map locally; additional tests cover the domain centre, halfway to the edge, edge ±1 and ±3 pulse SD, quarter-basal arc and antipode. Maximum absolute gain error is 1.69% of map amplitude. The spatial response supplies the drift prediction in G.

D. Mean-state activity on the same centred physical arc coordinate as C and E. Dotted lines locate half-contrast edges. Spring uses material positions multiplied by mean circumference.

E. Nearly all predicted drift (99.6–100%) is generated within two input correlation lengths of the two edges, suggesting lateral-edge fluctuations as the cellular readout. Curves show signed per-position contributions divided by total predicted diffusion; negative values remain. Rounded shares per edge/plateau interior/basal arc (%) are: Wave-pinning C20: 50.0/<0.1/<0.1; Wave-pinning C60: 49.9/0.1/<0.1; Holmes C20: 50.0/<0.1/<0.1; Holmes C60: 50.0/<0.1/<0.1; Spring C1: 50.0/0/0; Spring C3: 49.8/0.2/0.2; Goryachev C20: 50.0/0/<0.1; Goryachev C60: 50.0/0/<0.1. Interior and basal zones exclude edge zones; exact signed fractions are supplied with the source data.

F. Wave-pinning C20 and C60: the pulsed edge (solid) and opposite edge (dashed) move together and retain a permanent shift after withdrawal. Grey marks pulse-on; displacement is divided by pulse dose. Largest/smallest terminal edge-gain folds are Wave-pinning 0.701, Holmes 0.914, Spring 0.989, Goryachev 0.088. Enlargement suppresses susceptibility strongly in Wave-pinning and Goryachev, while Holmes and Spring change little. Other time courses and integrated responses are supplementary.

Figure 3. Enlargement slows both turning routes and selectively suppresses edge susceptibility · continued

G. A weak input anywhere shifts the domain by g(x) times dose; noise is many such inputs; drift follows from g and the input covariance; nothing is fitted. Shared-translation predicted versus measured diffusion is divided by each quantity's own smallest-size value at the same input (domain drift is also shown in Figure 4G). Grey is equality. Colours encode size; shapes encode f=0.0001, 0.0003, 0.001, 0.003, 0.005, 0.01 (circle, square, upward triangle, downward triangle, diamond, filled plus). Horizontal intervals are measured 95% bootstrap intervals. Wave-pinning's terminal gain fold gives $0.70^2 \approx 0.49$; the full spatial covariance predicts 0.490, versus 0.473 measured at f=0.001. The square relation is approximate because spatial shape and covariance also change. Goryachev fails quantitatively and retains its own scale. Absolute diffusion and the independent-edge candidate are supplementary.

H. The drift prediction implies a local size exponent, compared with measured heading-persistence α from Figure 4E. Predictions use the exact mean input of every observation; no f=0.001 extrapolation is used. The drift subset requires f≤0.01 and multiple-domain fraction≤0.05 throughout the fixed size neighbourhood. Native stationary-state and pulse-linearity gates must also pass. Grey is equality; vertical intervals are observed 95% bootstrap intervals; colours and shapes match G. 53 qualified pairs include all 15 Wave-pinning observations at f=0.003, 0.005 and 0.01. Its predicted α spans 2.48–2.88, above the size-independent-drift expectation of 2. Goryachev has no qualified pair. Every excluded condition and reason remains in the source data.

At the right edge, last tested linear peak amplitudes from the smallest to largest size are Wave-pinning 0.01024→0.00512 (0.5-fold), Holmes 0.0256→0.0256 (1-fold), Spring 0.0512→0.0512 (1-fold), and Goryachev 0.16384→0.65536 (4-fold). The next doubling exceeds 10% gain departure. Full brackets, all-size response curves and fixed-dose duration tests are in the methods and supplement.

Narrative points and conclusions

- Enlargement slows both turning routes: sliding of the incumbent under a boundary-straddling cue, and replacement by a challenger under a separated cue. At separated placements, half-turn time becomes insensitive to further distance while remaining strongly size-dependent.
- Enlargement selectively suppresses the terminal response of the lateral edges: largest/smallest gain is Wave-pinning 0.701, Holmes 0.914, Spring 0.989, Goryachev 0.088, over a threefold circumference contrast. Those two edge zones generate 99.6–100% of predicted domain drift. Strong-cue turning and weak susceptibility therefore probe different response properties.
- The same native susceptibility and input covariance predict physical domain drift without fitting. Wave-pinning has size-normalized diffusion 0.490 predicted and 0.473 measured at f=0.001. Across the qualified drift regime, its size-suppressed diffusion

Figure 3. Enlargement slows both turning routes and selectively suppresses edge susceptibility · continued

predicts α=2.48–2.88, above the drift-geometry value 2, which is now compared directly with Figure 4E. Goryachev's absolute drift disagreement remains visible.

Proposed measurements and experimental follow-up

In paired WGD−/WGD+ cells, vary photoactivation offset from a boundary-straddling cue to a separated cue. Measure activity-heading time to half-turn and image whether the incumbent slides or a new region takes over. WGD+ cells are predicted to turn more slowly through both routes.

At matched weak photoactivation dose, measure the permanent common displacement of both lateral edges of the leading activity region. WGD+ cells should show reduced displacement for Wave-pinning-like or Goryachev-like dynamics, with little size dependence for Holmes-like or Spring-like dynamics. The quoted model folds refer to a threefold circumference contrast, rather than one genome doubling.

Measure fluctuations at the lateral edges and within the active plateau separately. The edge-localized mechanism predicts that lateral-edge fluctuations, rather than plateau-interior fluctuations, account for heading drift; track both edges to separate common translation from width changes.

Scope, limitations and pending validation

Goryachev absolute drift prediction fails despite small deterministic-map validation error.

Goryachev has no qualified drift-regime alpha pair; all failed fixed-neighbourhood estimates and reasons remain explicit.

Linear-limit amplitudes are geometric-grid brackets; separated-cue plateau is an operational finite-data median.

Deterministic drift, total heading decay and empirical migration persistence remain distinct.

Model definitions and readouts

$$G(x_0) = \sum_j g_j h(x_j - x_0) \tag{3.1}$$

The main map is the standard Gaussian-pulse response per dose, not raw cell-input g.

$$c_j = g_j \sum_k g_k \int_0^\infty \overline{K_u(j,k;t,t+v)}^{\,t}\, \mathrm{d}v, \qquad D_s = \sum_j c_j \tag{3.2}$$

Figure 3. Enlargement slows both turning routes and selectively suppresses edge susceptibility · continued

Signed per-position contributions retain the exact native input covariance.

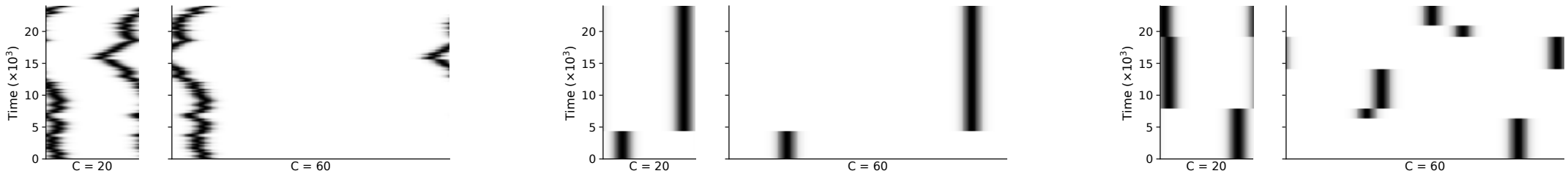
$$\alpha_{\text{pred}} = 2 - \frac{\mathrm{d}\ln D_s}{\mathrm{d}\ln \bar{C}} \tag{3.3}$$

Same fixed adjacent-size neighborhoods as Figure 4E; exact matched-input predictions and explicit drift-regime eligibility.

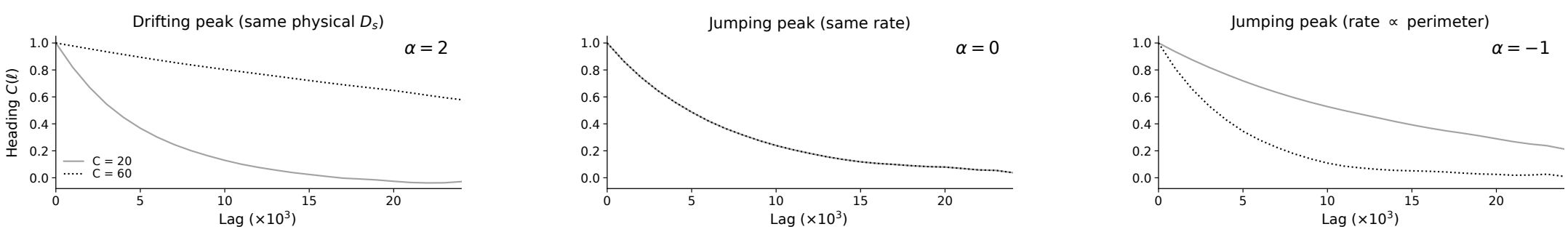
Sources and reproducibility

Scientific methods and unexpected results
Susceptibility maps and signed contributions
Every deterministic assay
Alpha bridge and unavailable rows

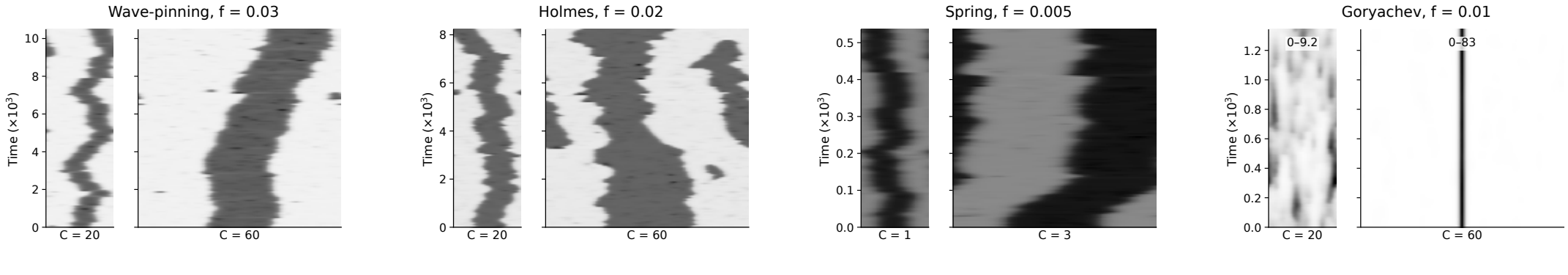
A


Drifting peak (same physical D_s)
Jumping peak (same rate)
Jumping peak (rate ∝ perimeter)
Time (×10³)
C = 20
C = 60

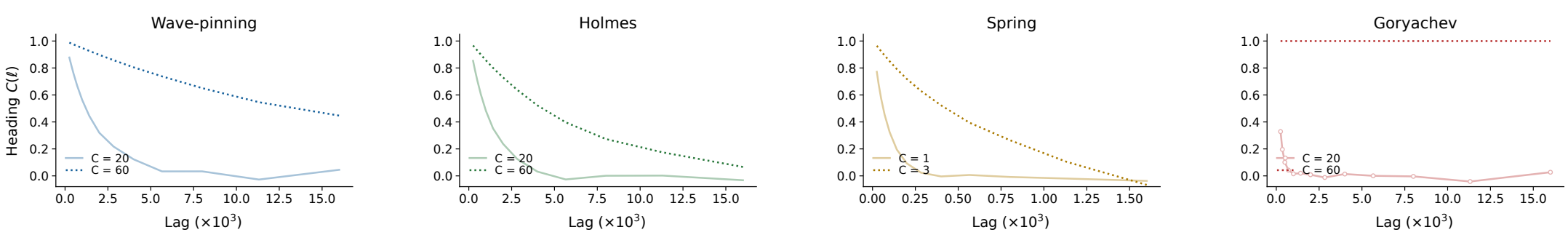
B


Drifting peak (same physical D_s)
Jumping peak (same rate)
Jumping peak (rate ∝ perimeter)
$\alpha = 2$
$\alpha = 0$
$\alpha = -1$
Heading $C(\ell)$
Lag (×10³)
C = 20
C = 60

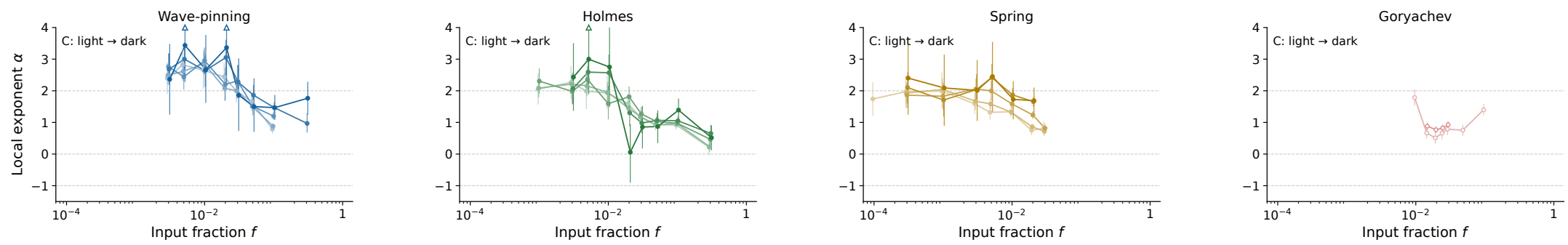
C


Wave-pinning, f = 0.03
Holmes, f = 0.02
Spring, f = 0.005
Goryachev, f = 0.01
Time (×10³)
C = 20
C = 60
C = 1
C = 3
0–9.2
0–83

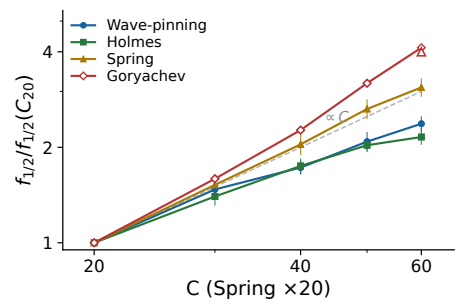
D


Wave-pinning
Holmes
Spring
Goryachev
Heading $C(\ell)$
Lag (×10³)
C = 20
C = 60
C = 1
C = 3

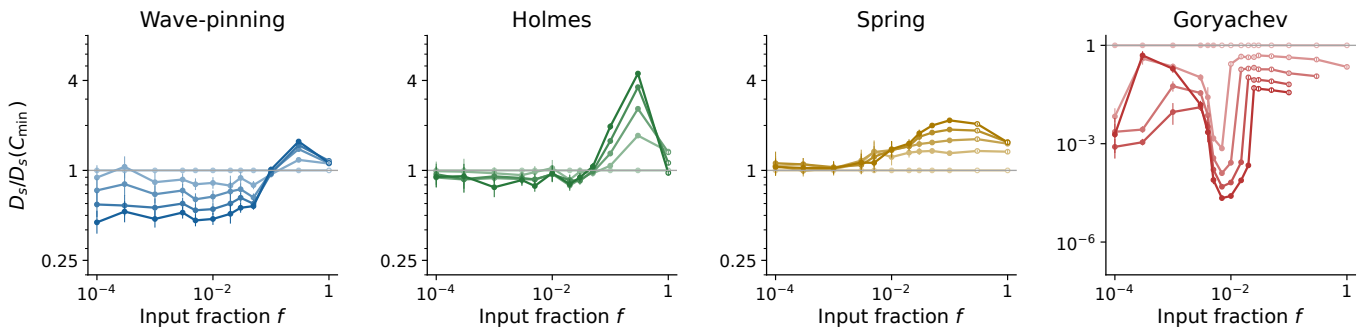
E


Wave-pinning
Holmes
Spring
Goryachev
C: light → dark
Local exponent α
Input fraction f

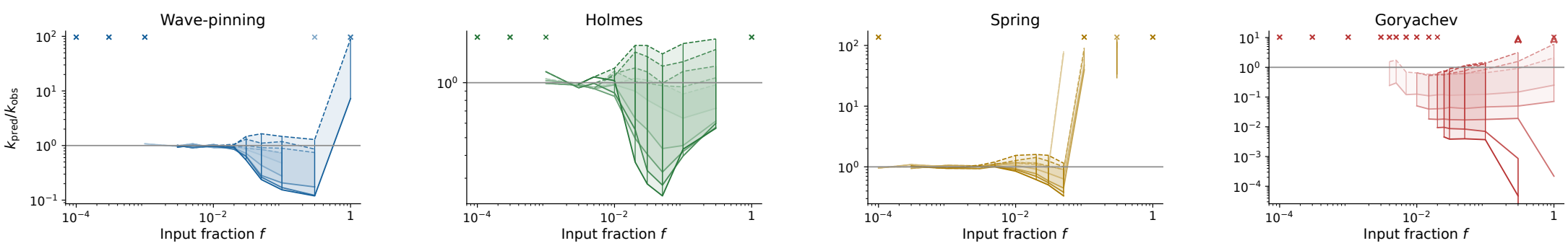
F


Wave-pinning
Holmes
Spring
Goryachev
$f_{1/2}/f_{1/2}(C_{20})$
C (Spring ×20)

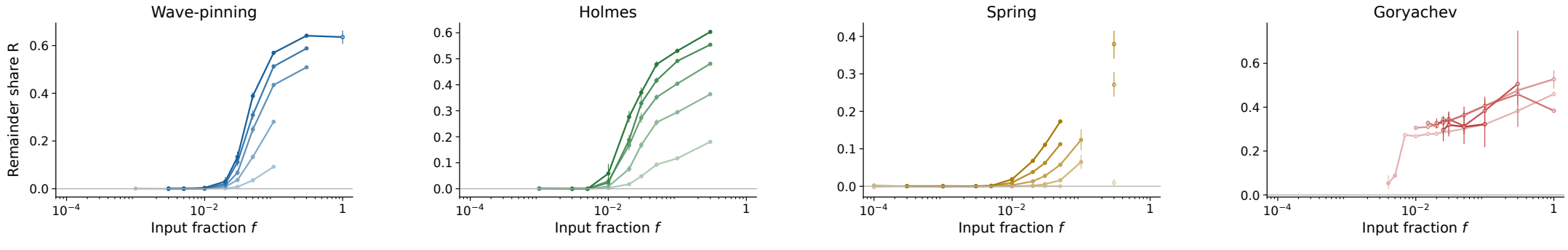
G


Wave-pinning
Holmes
Spring
Goryachev
$D_s/D_s(C_{min})$
Input fraction f

H


Wave-pinning
Holmes
Spring
Goryachev
k_{pred}/k_{obs}
Input fraction f

I


Wave-pinning
Holmes
Spring
Goryachev
Remainder share R
Input fraction f

J

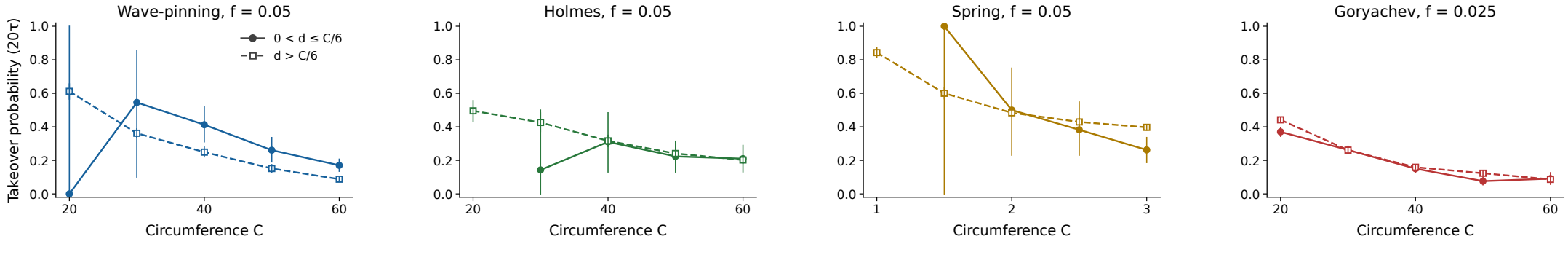

Wave-pinning, f = 0.05
Holmes, f = 0.05
Spring, f = 0.05
Goryachev, f = 0.025
0 < d ≤ C/6
d > C/6
Takeover probability (20τ)
Circumference C

Figure 4. Enlargement extends polarity heading memory under fluctuating input

Size-dependent heading persistence, region-based diagnostic limits and a detector-free mode test.

Detailed legend

Model panels use positive lognormal input with mean f times the native reference: Wave-pinning 0.1, Holmes 1, Spring 4 and Goryachev 6.4. The Gaussian driving field has unit variance, lognormal amplitude σ = 1, spatial parameter 1.5 and OU correlation time 50 (Spring basal-physical 0.15 and 5). Spring input is stationary in material coordinate, so its physical correlation length scales with circumference. Native clocks are not calibrated between models. Unless stated otherwise, error bars are 95% whole-history bootstrap intervals.

A, Detector-free null kymographs at C = 20 and 60: identical physical diffusion, size-independent jumping, or perimeter-proportional jumping. Each facet shows the same activity measure.

B, Null heading autocorrelation; persistence size exponents are 2, 0 and −1. C = 20 is light and C = 60 dark.

C, Native-model kymographs at f = 0.03 (Wave-pinning), 0.02 (Holmes), 0.005 (Spring) and 0.01 (Goryachev). Arc position is centred at the initial heading; time is elapsed after burn-in. Goryachev activity ranges are labelled separately for each size.

D, Model heading autocorrelation at the inputs in C, for the smallest and largest rings (Spring C = 1 and 3). Solid curves show the smallest ring and dotted curves the largest. Endpoint eligibility requires polarity coherence $|P| \geq 0.1$. These direct curves show longer heading memory in larger rings.

E, Local persistence exponent α versus input fraction f, using three neighbouring sizes (two at endpoints), with whole-history 95% intervals. In E and G–I, sizes are C = 20, 30, 40, 50 and 60 (Spring 1, 1.5, 2, 2.5 and 3), light to dark. Open circles in D, E, G and I mark the declared combined forcing-control flags; unavailable rates are omitted.

F, Noise-tolerance fold at heading-correlation lag 2000 (Spring 200), relative to the smallest ring. Spring circumference is multiplied by 20 for display. The dashed grey guide is proportional to circumference.

G, Physical within-tenure drift coefficient normalized by the smallest size at the same input, using five-frame persistent dominant tenures. Goryachev uses a different ordinate range, 10^{-7}–2; the other models use 0.2–8.

H, Drift plus counted five-frame-dominant takeovers, divided by observed heading decorrelation rate, at every designed input. Solid lower and dashed upper limits bound every unscored takeover loss between 0 and 2. Shading denotes identification intervals, not confidence intervals. Grey line: unity. Open upward triangles with arrows mark an unbounded upper limit where drift is unavailable. Only the unknown drift contribution

Figure 4. Enlargement extends polarity heading memory under fluctuating input · continued

has lower bound zero; counted-takeover lower contributions are retained. Crosses at the top mark unavailable observed rates.

I, Signed remainder share versus input, using point regions with native-amplitude scaling. It includes amplitude, within-region shape, unresolved activity and background contributions. The decomposition is descriptive; exact own-mask and normalization checks are supplementary.

J, Probability that a certified persistent secondary birth becomes globally dominant within 20 input correlation times. Filled circles: $0 < d \leq C/6$; open squares: $d > C/6$. Wave-pinning f = 0.05, minimum total births = 590; Holmes f = 0.05, minimum total births = 231; Spring f = 0.05, minimum total births = 580; Goryachev f = 0.025, minimum total births = 1064. For each model, select the designed input nearest in log distance to the smallest ring's half-memory input among inputs with at least 200 total certified births at every size, then drop $d \leq 0$. d denotes estimated edge-to-edge birth–incumbent separation, using the amount/global-contrast plateau-width approximation. Intervals resample whole histories 2000 times; [0,1] replaces intervals with fewer than 97.5% defined resamples. Positive-distance denominators and certification are reported separately. Lower 20-time-scale incidence can reflect slower or rarer takeover; lifetime takeover probability is unidentified.

The strong-input regime remains unexplained as a common four-model description. Rate-collapse screens pass in Wave-pinning, Holmes, Spring, Goryachev; local-slope screens pass in Wave-pinning, Holmes, Spring. This partial agreement depends on the local-slope stencil: Wave-pinning and Spring fail the three-point slope screen, so it does not establish a robust strong-regime explanation. Fourier-mode tests (K–L), spectra and exact-mask diagnostics appear in the supplement.

Narrative points and conclusions

- Larger rings retain heading longer near the transition; the local size exponent decreases as input strengthens. The largest-to-smallest noise-tolerance folds are 2.37 (Wave-pinning), 2.16 (Holmes), 3.09 (Spring) and 4.12 (Goryachev).
- Physical weak-input drift decreases with size in Wave-pinning but is nearly size independent in Holmes and Spring. The deterministic drift bridge is quantitatively unreliable for Goryachev. Region-based closure and remainder delimit the inputs where this description is adequate.
- At each model's selected input, far-born certified fronts have lower finite-horizon takeover estimates in larger rings; near-bin estimates need not decline monotonically. Certification and finite follow-up limit the population and lifetime interpretation.

Figure 4. Enlargement extends polarity heading memory under fluctuating input · continued

- Fourier rate-collapse screens pass in Wave-pinning, Holmes, Spring, Goryachev; local-slope screens pass in Wave-pinning, Holmes, Spring. This partial agreement depends on the local-slope stencil: Wave-pinning and Spring fail the three-point screen. It does not establish a robust strong-regime explanation. All three/five/eight-point metrics are reported; this is a descriptive test, not causal attribution.

Proposed measurements and experimental follow-up

In matched WGD− and WGD+ cells under the same fluctuating input, image full-perimeter polarity activity and estimate heading autocorrelation jointly with coherence. Longer heading memory in enlarged cells is the prediction; this readout is distinct from velocity persistence.

Track the physical lateral displacement of both front boundaries. Size-suppressed physical drift would favour a Wave-pinning-like susceptibility change; nearly size-independent physical drift with slower angular wandering would favour Holmes/Spring-like geometry. The Goryachev quantitative bridge is unreliable.

Count persistent secondary protrusion births per perimeter–time and follow new-front dominance conditional on birth distance, using both 20 input correlation times and longer follow-up. The prospective signature is lower far-born takeover incidence with enlargement; near-born signatures are model dependent. Distinguish births from splits and retain censored histories to separate slow resolution from rare takeover.

Compare photoactivation that straddles the existing front boundary with separated activation, and measure time to half-turn across cell sizes. This tests the two imposed-turning routes and whether the size effect changes with cue position.

Measure perimeter Fourier spectra and mode correlations against physical wavenumber in both ploidy groups. A common rate curve would support a detector-free description; disagreement would reject that description without requiring a domain detector.

Scope, limitations and pending validation

Missing takeover angles are bounds, not imputed means.

Exact birth/split population classification remains partly unidentifiable.

Two exemplar seeds cannot validate universal mode collapse.

A universal strong-input mechanism is not identified.

Model definitions and readouts

Figure 4. Enlargement extends polarity heading memory under fluctuating input · continued

$$Q_t = \frac{\sum_{i \in I_t} A_{i,t} e^{i\theta_{i,t}}}{\sum_{i \in I_t} A_{i,t}}, \qquad r_t = |P_t|, \quad \widetilde{Q}_t = r_t Q_t \tag{4.1}$$

Q is the normalized point-region moment; P=M₁/M₀ is measured from the native field. Its amplitude r prevents the unit-modulus single-region normalization artefact.

$$\Delta P_j = \bar{r}\,\Delta Q_j \ (j = \mathrm{M}, \mathrm{A}, \mathrm{B}), \qquad \Delta P_{\mathrm{R}} = \Delta P - \bar{r}\,\Delta Q \tag{4.2}$$

The three ΔQ components use the symmetric Shapley convention for motion, coexisting amount and membership; r̄ is mean endpoint native amplitude. R closes the native moment change and includes amplitude variation, shape and background. This is an accounting convention, not causal attribution.

$$L_j = \frac{\mathrm{Re}\left[\Delta P_j \overline{(H_1 - H_0)}\right]}{|P_0| + |P_1|}, \quad H_t = \frac{P_t}{|P_t|}, \quad \sum_j L_j = 1 - \cos(\theta_1 - \theta_0) \tag{4.3}$$

Pooled signed fractions are sums of L_j divided by total measured heading loss. They retain cancellation, can exceed one, and depend on the stated accounting convention; they are not causal shares.

$$k(\ell) = -\frac{\ln C(\ell)}{\ell}, \qquad k \propto \bar{C}^{-\alpha} \tag{4.4}$$

C(ℓ) is activity-heading autocorrelation; mean C is actual circumference. Endpoint eligibility uses |P|≥0.1. Main H evaluates every designed input; unavailable observed rates remain explicit.

$$a_n(t) = \frac{1}{N} \sum_{j=1}^{N} u_j(t) e^{in\theta_j}, \qquad P(t) = \frac{a_1(t)}{a_0(t)}, \qquad q_n = \frac{2\pi n}{\bar{C}} \tag{4.5}$$

Fourier modes use cell-centre material phases and the positive convention matching the native first moment. Heading is the phase of P; Spring physical wavenumber uses mean circumference.

$$\gamma_n = -\frac{\ln C_n(\ell_*)}{\ell_*}, \qquad k(C) = \gamma(2\pi/C), \qquad \alpha = \frac{\mathrm{d}\ln\gamma}{\mathrm{d}\ln q} \tag{4.6}$$

The last two equalities are the detector-free hypothesis under a common size-independent mode curve, not an identity assumed in analysis. Complex correlations use the nearest-half lag rule. Half-crossing status is exported; primary collapse and slope comparisons use observed crossings. The declared descriptive screens and stencil sensitivities are reported model by model.

Figure 4. Enlargement extends polarity heading memory under fluctuating input · continued

$$k^{\mathrm{lo}}_{\mathrm{pred}} = q_1^2 D_s + \frac{\sum_{\mathrm{scored}} (1-\cos\Delta\theta)}{T}, \qquad k^{\mathrm{hi}}_{\mathrm{pred}} = k^{\mathrm{lo}}_{\mathrm{pred}} + \frac{2N_{\mathrm{unscored}}}{T} \tag{4.7}$$

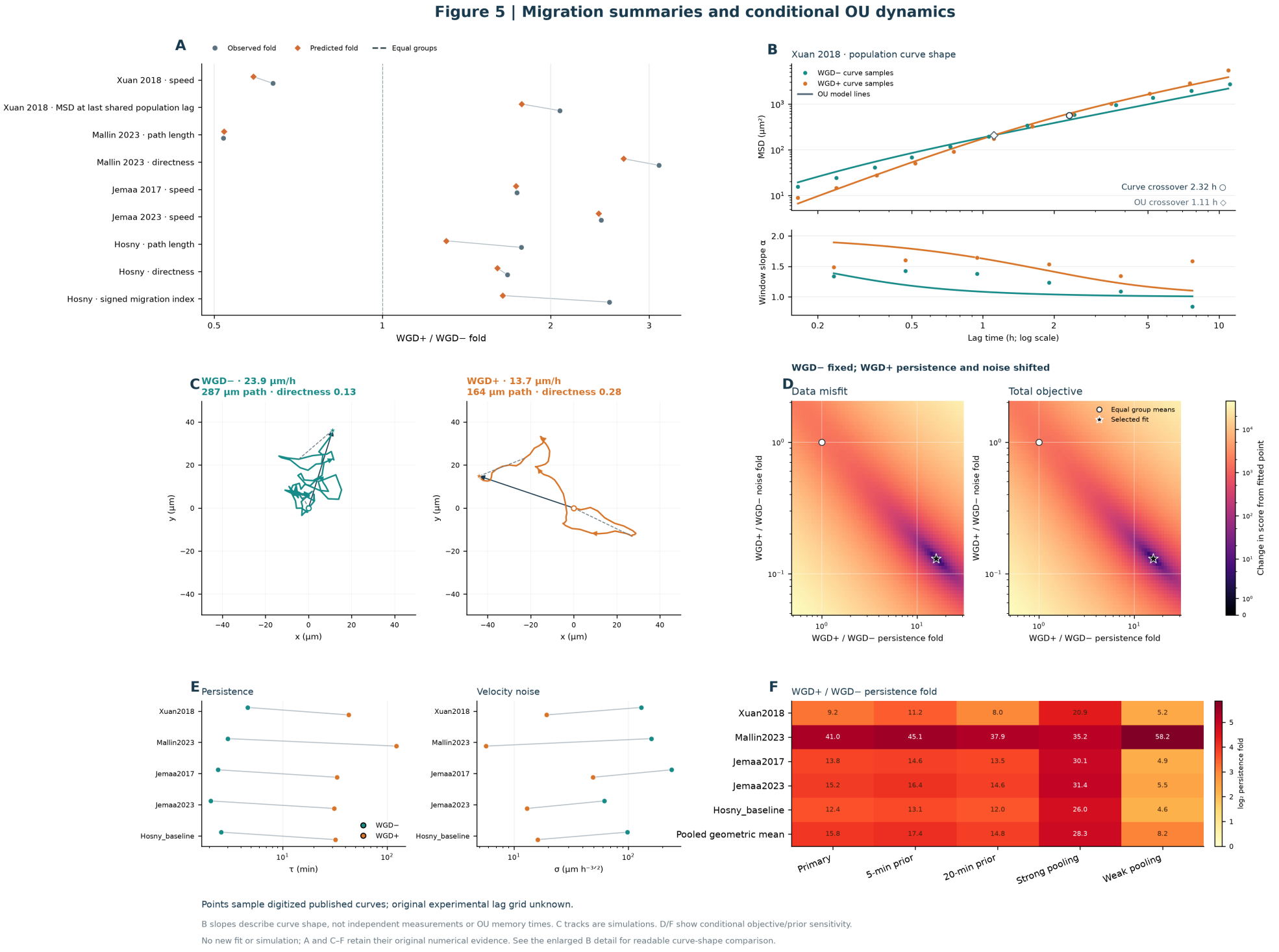
All counted takeovers enter the interval: each unscored angular loss lies between 0 and 2. T is eligible exposure. Missing drift has a nonnegative lower bound and unbounded upper limit; an unavailable observed heading rate prevents a numerical ratio.

Sources and reproducibility

methods provenance.md

scientific readme.md

Figure 5. Migration observations and conditional OU interpretation · image 1 of 2



Reassembled Figure 5: original A and C–F numerical evidence, revised MSD/curve-shape panel B.

Figure 5. Migration observations and conditional OU interpretation · image 2 of 2

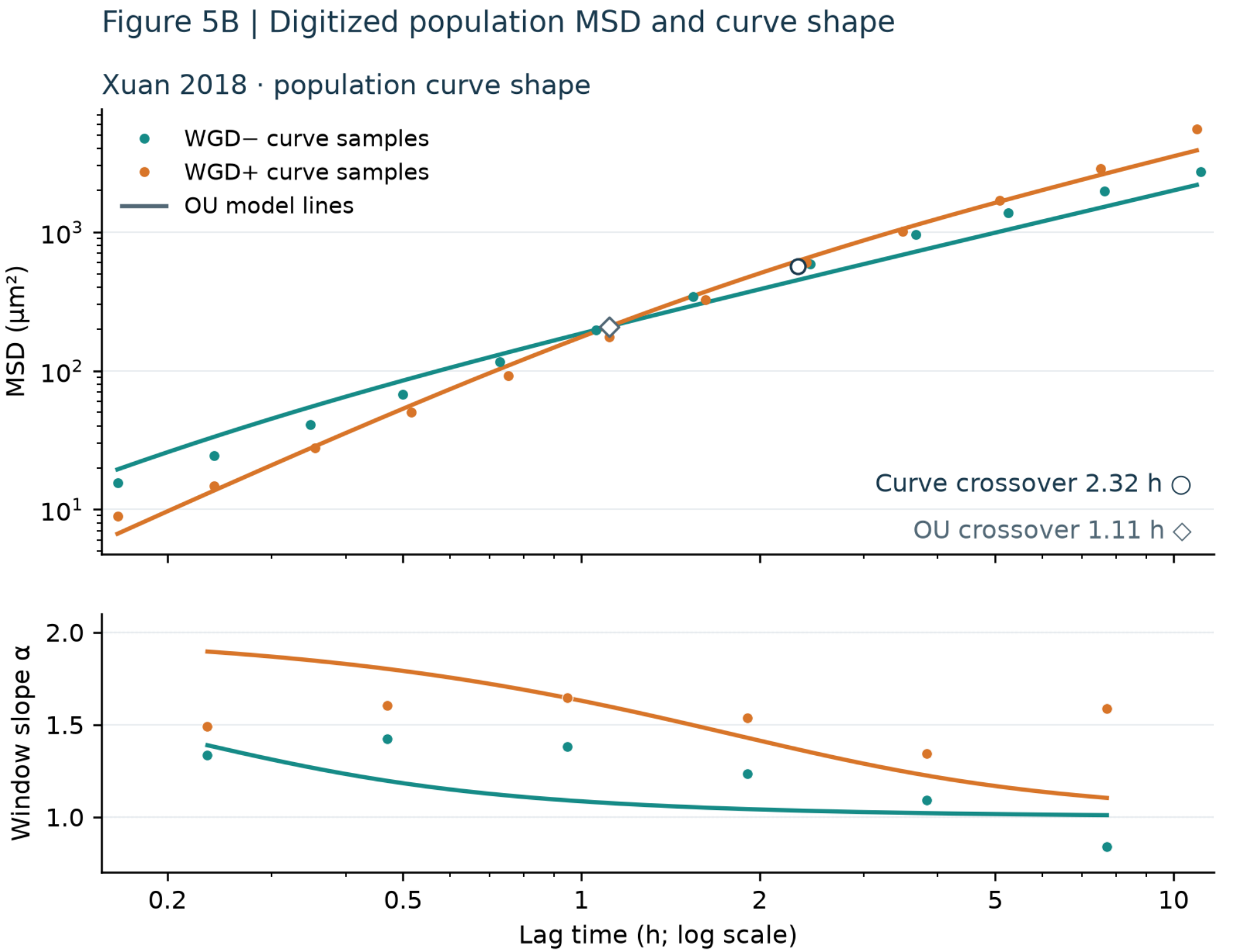


Figure 5B continued: enlarged detail with explicit raster-sample caveat.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Existing empirical support, with observed migration summaries kept separate from fitted velocity-memory parameters.

Detailed legend

A, Published/digitized enlarged or polyploid-group to reference-group ratios for nine migration summaries, alongside primary Ornstein–Uhlenbeck (OU) fit predictions. Xuan contributes sampled mean speed and population mean-squared displacement (MSD) at the last shared lag, approximately 10.95 h; Hosny summaries are group medians. Mallin central-bar statistic is not specified by the source and retains the historical mean approximation. These distinct source summaries are not interchangeable measures of persistence.

B, Upper: twelve chosen display samples from each digitized Xuan population MSD curve (small points) and the unchanged fitted analytical OU expectation (solid lines), on logarithmic lag and MSD axes. These are samples of the published raster curves, not recovered experimental lag positions or independent cell observations; the original experimental lag grid is unknown. The open circle identifies the crossover of the full digitized traces at approximately 2.32 h, computed by linear interpolation in native lag and MSD. The open diamond identifies the different fitted OU crossover at approximately 1.11 h. Lower: descriptive slopes of log(MSD) versus log(lag) fitted in six equal log-lag windows of the full digitized curves (points), and matching-window moving OU slopes (solid lines). Each window spans approximately 0.304 decades; both observed and model slopes use the same full window support. Observed raster columns remain correlated and give no biological sample size or confidence interval. These slopes are newly calculated curve-shape summaries, not author-reported local exponents or fitted OU memory times. A constant power-law exponent would appear horizontal; reference values 1 and 2 indicate diffusive and straight-line scaling. The model does not reproduce the complete observed curvature or the observed crossover. The analytical fit, all fitting targets and numerical inputs to A and C–F are unchanged.

C, Representative Xuan primary-fit model-generated tracks, with sampled speed, total path length and directness (net displacement divided by path length). Both groups use identical spatial scales. The tracks are simulations, not published individual-cell trajectories; a slower sampled trajectory can nevertheless have greater endpoint directness. Dashed chords illustrate the contribution of finite-time displacements.

D, With reference-group parameters fixed, vary enlarged-group persistence and velocity-noise ratios. Separate surfaces show change in the descriptive data-misfit score and in the total regularized objective; the selected fit and equal-group reference are marked. The heuristic score is not a calibrated likelihood, and the surfaces are not confidence regions.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

E, Primary paired persistence and velocity-noise estimates for Xuan 2018, Mallin 2023, Jemaà 2017, Jemaà 2023 and the Hosny hypoxic baseline. F, Enlarged/reference persistence folds under the primary 10-minute prior, 5- and 20-minute alternatives and stronger/weaker pooling, with the pooled geometric mean. Priors and pooling define conditional sensitivity scenarios, not uncertainty intervals. The Jemaà comparisons constrain speed alone, leaving persistence/noise tradeoffs dependent on hierarchy and priors.

The fitted population contrasts span stable tetraploids, giant polyploids and treatment-induced polyaneuploid cancer cells (PACCs). The inherited WGD+/WGD− labels in the artwork are comparison shorthand and do not establish matched WGD-only interventions in every study. Hosny observations derive from the recovered preprint; equivalence to the journal version remains unresolved.

Narrative points and conclusions

- Xuan combines lower sampled speed (0.64-fold) with greater late-lag population MSD (2.08-fold); Mallin combines shorter path length (0.52-fold) with greater directness (3.12-fold). Persistence-related behaviour therefore deserves analysis separately from speed.
- Hosny baseline PACCs have greater median path length (1.77-fold) and directness (1.67-fold). Signed migration index is a separate directional summary.
- All five primary OU comparisons infer longer velocity memory. The pooled primary persistence fold is about 15.8, and remains above one across the declared prior/pooling scenarios; this is a conditional model result.
- The fit does not reproduce the complete Xuan curve shape within its raster envelope, and data-only sensitivity has rank 18 of 22. Optimization completion does not establish model adequacy or uniquely identified persistence times.
- The observed Xuan curves cross near 2.32 h while the unchanged fitted OU curves cross near 1.11 h. Matched-window slopes make the remaining curve-shape discrepancy visible; fitted persistence is a conditional summary.
- Lower sampled speed can coexist with greater late-lag displacement in the enlarged-group comparison. Its causal connection to polarity mechanisms requires matched activity and migration measurements.

Proposed measurements and experimental follow-up

The new WGD assay should measure matched trajectories directly, with declared sampling and observation duration, and assess speed, displacement, directness and lag-dependent MSD before considering fitted persistence. An activity-polarity memory

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

measurement would complement these quantities without equating activity and velocity.

Scope, limitations and pending validation

Neither these heterogeneous group comparisons nor the OU fits establish that enlargement alone causes persistence. Cell identity, induction/treatment and acquisition differences require explicit treatment in the new experiments.

The requested observed-point/model-line and curve-shape display is implemented. The original experimental lag grid remains unknown; neither display samples nor raster-derived slopes supply independent biological replicates. The observed/model crossover discrepancy remains visible.

Sources and reproducibility

Refined OU analysis
Original panel legend
Observed folds
Fit results and adequacy
Hosny sensitivity supplement
Revised display, exact preserved inputs and direct-user feedback
Migration display numerical and preservation checks

Figure 6. RESERVED: new WGD motility results

RESERVED MAJOR EXPERIMENTAL RESULTS

New WGD motility assays — all data pending

A Paired biological system

SUM-159 chemically induced WGD+/− pilot

Confirm group identity, size and morphology

PENDING — no observations available

B Imaging and observed tracks

Optimize acquisition and tracking

Declare cadence, duration, eligibility and replicates

PENDING — no observations available

C Measured migration behaviour

Speed, path length, displacement and directness

Lag-dependent MSD under declared observation rules

PENDING — no observations available

D Evaluate the persistence hypothesis

Relate the acquired observations to WGD and size

Retain null or opposite outcomes; specify controls

PENDING — no observations available

Proposed panel plan only. Design and analysis remain to be agreed before interpreting data.

Explicit placeholder. Every panel is pending acquisition, analysis and author review; no mock observations are displayed.

Figure 6. RESERVED: new WGD motility-assay results

A substantial new experimental results section and main figure. No observations are available; the panel layout is a proposal, not an experimental result.

Detailed legend

A, Proposed experimental context: the SUM-159 chemically induced WGD+/− pair is the agreed starting pilot with the Speros collaboration. Imaging should first be optimized for that pair before expansion to additional cell lines. Ploidy/group identity, physical size and morphology need to be documented so the intended comparison is clear.

B, Reserved for representative experimental fields and trajectories under the selected acquisition conditions. The eventual panel should state sampling interval, usable duration, spatial scale, track eligibility and exclusions. Number of independent preparations and biological replicates remains to be specified.

C, Reserved for empirical motility summaries: sampled speed, accumulated path length, endpoint displacement/directness and lag-dependent MSD. The final readouts, follow-up window and analysis plan are to be settled before interpreting the data; this placeholder does not imply that all listed measurements will be technically feasible.

D, Reserved for the measured relationship between WGD-associated enlargement and migration behaviour, including size distributions and controls needed to interpret the group contrast. Any activity/protrusion measurements arising from the mechanism proposals require a separately agreed assay and should be distinguished from the basic motility pilot. Statistical models, uncertainty reporting and sample-size targets remain open.

Narrative points and conclusions

- This major results section will test whether the planned paired experimental system exhibits the migration behaviour motivating the theoretical work.
- Its conclusions must follow the acquired observations, including null or opposing results. Simulation output and literature fits cannot fill this gap.
- Agreement between increased polarity resistance or activity memory and migration persistence would support the proposed connection; causal attribution requires appropriate perturbations and controls.

Proposed measurements and experimental follow-up

Pending decisions include plating density, acquisition duration and cadence, field/condition throughput, tracking and exclusion rules, replicate structure, confirmation of WGD status, size quantification and the statistical analysis. The known throughput versus spatial/temporal resolution trade-off should be resolved by the pilot.

Figure 6. RESERVED: new WGD motility-assay results · continued

Mechanism validation can subsequently examine incumbent extent, early challenger growth, local loss/rebound, site fates and global activity-polarity correlation under characterized perturbations. These activity experiments are proposals alongside the theoretical mechanisms and are separate from the reserved motility-results contribution.

Scope, limitations and pending validation

No pilot acquisition, sample size, effect estimate, significance claim or experimental outcome is supplied by the available evidence.

Sources and reproducibility

Author manuscript vision and supplied collaboration context

Supplementary material and supporting analyses

Figure 1: round6 complete supporting evidence

Current source data, numerical qualification and preceding supplementary evidence; earlier main artwork is preserved.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round6_20261023/figure1/figure1_round6_supplement.pdf

Scientific results and rebuild instructions

Figure 2: round6 complete supporting evidence

Current source data, numerical qualification and preceding supplementary evidence; earlier main artwork is preserved.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round6_20261023/figure2/figure2_round6_supplement.pdf

Scientific results and rebuild instructions

Figure 3: round6 complete supporting evidence

Current source data, numerical qualification and preceding supplementary evidence; earlier main artwork is preserved.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round6_20261023/figure3/figure3_round6_supplement.pdf

Scientific results and rebuild instructions

Figure 4: round6 complete supporting evidence

Current source data, numerical qualification and preceding supplementary evidence; earlier main artwork is preserved.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round6_20261023/figure4/figure4_supplement.pdf

Scientific results and rebuild instructions

Prospective SUM-159 WGD pilot observables

Specified on 23 October 2026, before SUM-159 WGD+/− motility data; no experimental observations were used.

Drift classification: f ≤ 0.01 and multiple-domain fraction ≤ 0.05; Figure 4H shows all inputs. In cells classify coexisting protrusions; native f and time are uncalibrated.

Cellular observable	Model counterpart	Drift regime: WGD+ / WGD−	Secondary-domain regime: WGD+ / WGD−	Mechanistic interpretation
Common-mode motion of the leading region's two lateral edges	Correlated boundary translation and domain drift (Figs 3F–G, 4G)	Similar (Holmes/Spring); smaller in Wave-pinning/Goryachev	No common direction identified outside persistent single-domain bouts	Translation; does not identify a Figure 2 route
Secondary-protrusion birth rate per unit perimeter	Persistent no-overlap birth rate (Fig. 4 supplement: certified births)	Rare in both groups	Model dependent: Spring higher and Goryachev lower at f = 0.1; other trends unresolved	Initiation; model-dependent size response
Takeover incidence within a fixed follow-up window	Certified takeover incidence (Fig. 4J; pooled 80τ sensitivity in supplement)	Not applicable when births are too rare	Lower by 20τ; persists through 80τ in the same cohort; lifetime fate unidentified	Incumbent resistance; lifetime success unresolved
Activity/protrusion-heading autocorrelation time	Heading decay, size exponent and tolerance (Fig. 4D–F)	Longer; approximately size squared only when physical drift is size independent	Size advantage shrinks in the tested models; compare coherence and domain counts	Regime readout; no unique route assignment
Edge displacement after boundary-straddling photoactivation	Weak deterministic spatial and edge response (Fig. 3C, F)	Smaller (Wave-pinning/Goryachev); similar (Holmes/Spring)	Not specified: the pulse assay uses a stable polarized incumbent	Susceptibility within a route, not a unique mechanism

Prospective directions, not SUM-159 observations or a full statistical preregistration. Freeze sampling, segmentation, physical pulse dose and group inference before comparison; model f and native times have no calibrated cellular equivalents.

Birth rates are model dependent: at f = 0.1, bounds identify a Spring increase and a Goryachev decrease with size. Takeover within 20τ is fixed-horizon incidence; its decline persists through 80τ in the same cohort, while lifetime fate and certification selection remain unresolved.

Common translation tests the drift mechanism. Edge pulses refine susceptibility within routes: Wave-pinning and Holmes share an expansion-constrained class in Figure 2 but predict different edge responses. Combine these readouts with Figure 2 first-rise, width and pulse-gap assays to test initiation-limited reset, saturation/expansion and retained progress after visible activity loss. Activity heading, observed migration and OU-fitted memory remain separate readouts.

The takeover direction summarizes the pooled-input common cohort. Figure 4J instead separates distance at one input per model; its probabilities need not decrease monotonically in every size bin. The two comparisons have different conditioning and must not be combined.

Prospective pilot observables before SUM-159 data

Specified on 23 October 2026, before SUM-159 WGD+/− motility data; no experimental observations were used.

Detailed legend

Prospective directions, not SUM-159 observations or a full statistical preregistration. Freeze sampling, segmentation, physical pulse dose and group inference before comparison; model f and native times have no calibrated cellular equivalents.

Birth rates are model dependent: at f = 0.1, bounds identify a Spring increase and a Goryachev decrease with size. Takeover within 20τ is fixed-horizon incidence; its decline persists through 80τ in the same cohort, while lifetime fate and certification selection remain unresolved.

Common translation tests the drift mechanism. Edge pulses refine susceptibility within routes: Wave-pinning and Holmes share an expansion-constrained class in Figure 2 but predict different edge responses. Combine these readouts with Figure 2 first-rise, width and pulse-gap assays to test initiation-limited reset, saturation/expansion and retained progress after visible activity loss. Activity heading, observed migration and OU-fitted memory remain separate readouts.

Narrative points and conclusions

- Model drift subset: f ≤ 0.01 and multiple-domain fraction ≤ 0.05. In cells, classify by the observed frequency of coexisting protrusions; model f and native times have no calibrated cellular equivalents. This gate defines the drift interpretation; Figure 4H displays closure at all inputs, including outside this subset.

Proposed measurements and experimental follow-up

Track both lateral half-contrast boundaries of a persistent leading activity/protrusive region. Report their mean-squared physical displacements and cross-correlation over declared lags, including failed single-domain eligibility and changes in region identity. Compare physical displacement before normalizing to perimeter. The two edges are strongly correlated in model exemplars, so report their covariance rather than assuming independence. Report common-mode displacement (left plus right)/2 separately from width fluctuations (right minus left); a single-edge MSD alone cannot distinguish these modes.

With full images retained, call a new connected protrusive/activity region a birth only if it overlaps no region in the preceding frame and persists for at least five consecutive saved frames. Divide by observed perimeter-time exposure; count splits separately. Freeze segmentation, cadence and overlap rules before group comparison.

For each qualifying birth, measure shortest centroid arc distance from the incumbent minus the incumbent half-width, estimated as excess amount divided by twice the

Prospective pilot observables before SUM-159 data · continued

declared plateau height. A takeover requires five consecutive frames of dominance. Report right censoring and a common follow-up horizon; distinguish all births from any subset whose identity is certifiable. The main model comparison uses a common 20-noise-correlation-time follow-up at one declared input per model, with at least 200 certified births at every size. Pooled inputs and the 80τ common-cohort sensitivity are supplementary; these conditioning choices must be separated from the cellular analysis. This is fixed-horizon incidence, not a lifetime takeover probability. Record time since birth, cumulative incidence and right censoring. In the same model cohort with at least 80τ follow-up, the decline persists from 20τ to 80τ; finite observation and certification selection still prevent a population lifetime claim.

Define the activity/protrusion vector from the full spatial signal; estimate heading autocorrelation at declared lags with a fixed coherence rule and report excluded pairs. Compare groups at matched analysis lags and observation duration. Keep cell displacement/velocity persistence as a separate observed readout.

Apply a weak, brief photoactivation spot centred on a lateral leading-edge boundary, matched in physical width and duration across ploidy groups. Track both edges through relaxation, subtract a no-pulse control, and check half/double amplitudes. A response supports a mechanism only to the extent that its size trend differs among model predictions. At model f = 0.001, largest/smallest pulse-response gains are 0.701 (Wave-pinning), 0.914 (Holmes), 0.989 (Spring), and 0.0878 (Goryachev); these are native-model size contrasts, not numerical cellular fold predictions. Separately repeat the Figure 3A–B assay with cue placements straddling and separated from the incumbent boundary, measuring half-turn time to distinguish sliding from takeover; the weak local pulse and the finite turning cue are distinct assays.

Scope, limitations and pending validation

No SUM-159 motility data are available in this revision. Directions are prospective and must not be described as confirmed experiments.

Native model times and f are not calibrated to cells. Event holds measured in frames depend on acquisition cadence; select and report that cadence before comparing experimental groups.

The compact model records do not identify every birth, split or local plateau height. Model bounds and certified-subset probabilities do not establish the unconditional cellular birth/takeover directions.

The new observables do not uniquely distinguish the Figure 2 response classes. Weak boundary susceptibility and finite-cue sliding/takeover are different properties; the within-20τ size contrast cannot distinguish slower completion from a change in eventual success.

Prospective pilot observables before SUM-159 data · continued

Sources and reproducibility

Prospective directions and operational definitions

Supplementary material and supporting analyses

Figure 3: edge-response / secondary-domain revision supplement

New measurements, identification limits, numerical checks and preceding supplementary evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round5_20261023/figure3/figure3_round5_review_supplement.pdf

Physical diffusion prediction and observed drift

Size-normalized prediction and observed drift

Normalized integrated boundary response

Methods and covariance derivation

Both boundary assays and numerical checks

Reviewed main/supplement preservation

Figure 4: edge-response / secondary-domain revision supplement

New measurements, identification limits, numerical checks and preceding supplementary evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round5_20261023/figure4/figure4_supplement.pdf

Round5 complete campaign analysis

Scientific identity/enclosure validation

Birth and heading-loss conditions

Takeovers with common follow-up

Original campaign manifest

Native-amplitude heading accounting

Native-amplitude condition metrics

Figure 4: corrected event scoring and full closure comparisons

Scorable-event coverage, missing-angle bounds, both detector-free cadences and all closure failures. Preceding editions remain linked below.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round4_20261023/figure4/figure4_supplement.pdf

Scientific methods and regime-specific counts

Figure 4: round-3 analysis and null comparisons

Pointwise and smooth size exponents, drift/jump cross-checks, sensitivity, valid fractions, replication and analytic null checks. Missing estimates and finite-rate precision shortfalls are identified in the scientific readme.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure4/figure4_supplement.pdf

Unexpected results, data status and methods

Supplementary material and supporting analyses · continued

Figure 2: round-3 evidence and preserved prior panels

The complete supplement adds the per-width replacement curves and the deBelly native endpoint audit, then preserves the complete round-2 main and every earlier supplement page. Superseded fate maps, ratios and class tables are historical evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure2/figure2_round3_supplement.pdf

Round-3 reader and source data

Exact preservation receipt

Figure 3: round-3 evidence and every preserved round-2 panel

The supplement retains 90/99%-heading time curves, the preceding four-placement round-3 main and legend including C = 47.6 profiles, and the complete round-2 main and supplement. Original source files remain unchanged.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure3_single_C/figure3_round3_supplement.pdf

Round-3 reader and native source coordinates

Exact supplement preservation

Historical Figure 4: previous supplement and decision record

Historical evidence from the preceding Figure 4 edition, preserved unchanged. The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history. S0–S12 statistical core and complete native-atlas appendix index; native pages are in all linked appendix volumes. Every numbered appendix below is part of the complete supplement.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/signature_figure4_20261020/supporting/vectors/figure4_revision_complete_supplement.pdf

Figure reader and source data

Independent completion review

S11 native atlas appendix 001

S11 native atlas appendix 002

S11 native atlas appendix 003

S11 native atlas appendix 004

S11 native atlas appendix 005

S11 native atlas appendix 006

S11 native atlas appendix 007

S11 native atlas appendix 008

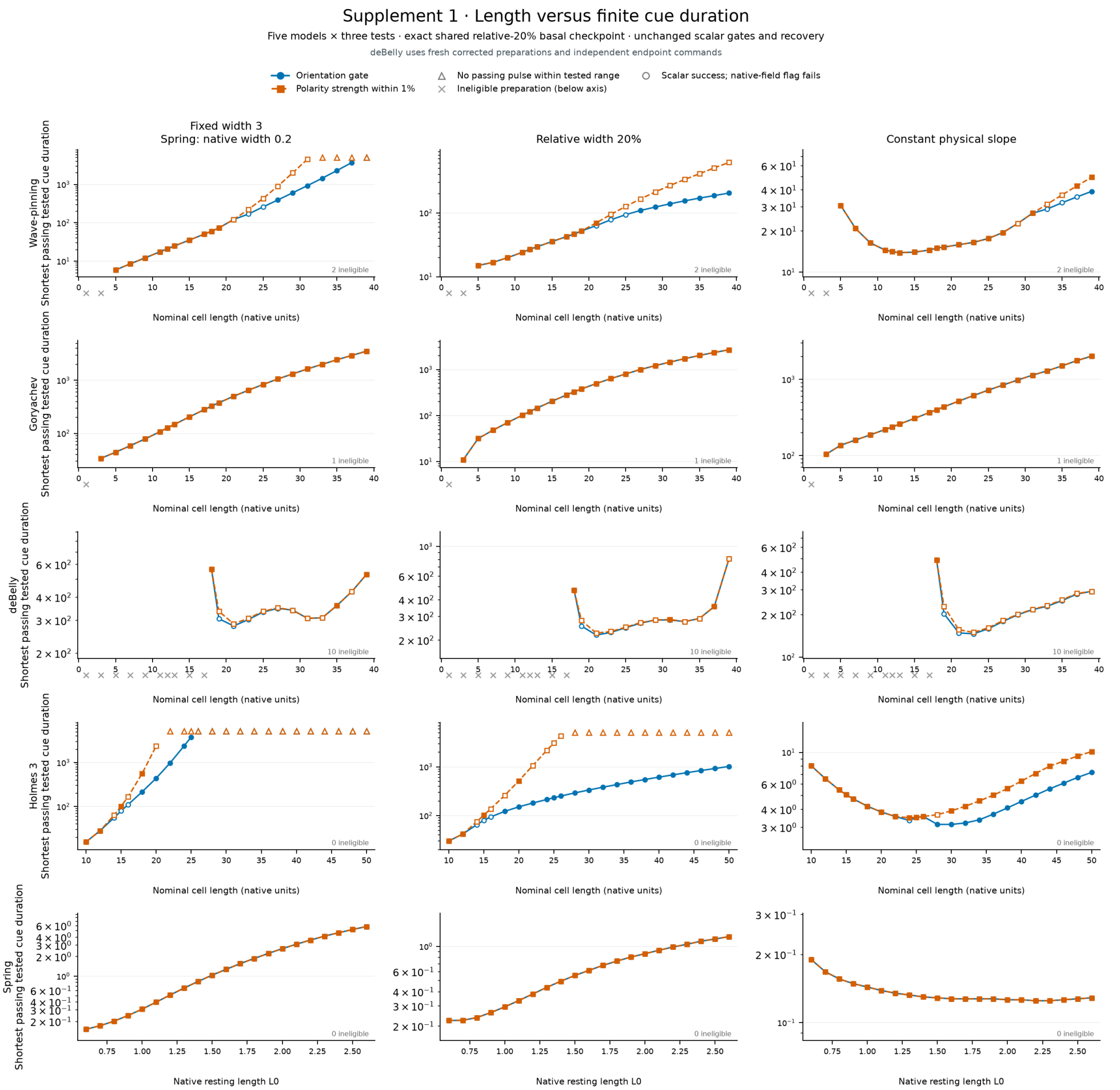
S11 native atlas appendix 009

S11 native atlas appendix 010

Supplementary material and supporting analyses · continued

S11 native atlas appendix 011
S11 native atlas appendix 012
S11 native atlas appendix 013

S1. Full cue-duration response curves



Both patches use a constant peak across all lengths; Goryachev K=64, peak=6.4. Gradients retain one constant physical slope per model.
Each positive duration restarts the original checkpoint. Full strength adds the 1% reversed initial-polarity band to the orientation gate.
Open markers retain failing field-tail flags. Censoring marks explicit tested limits; missing endpoints break lines. Native times and scales differ.
Data generated 2026-10-08T19:26:33.200718+00:00; producer finished. Every imported and fresh trial passed independent native-array audit.

Canonical corrected five-model duration curves; source working number Supplement 1.

Earlier supporting material: S1. Full cue-duration response curves

Essential curve context for the main comparison heatmap.

Detailed legend

Five model rows and three finite-cue columns retain the complete sampled length dependence. Coloured curves distinguish orientation from strength replacement. Open success markers retain scalar successes that fail the separate native-field tail flag; triangles indicate no passing pulse within the tested range; crosses indicate ineligible preparations. Missing values are not zero response times. Each duration excludes preparation and the subsequent 10,000-unit basal recovery. Spring uses native length and time. See Figure 1 for exact gates and stimulus conventions.

Narrative points and conclusions

- Corrected deBelly is nonmonotone across all three geometries. The positive rank coefficients summarize association rather than its full curve shape.
- Constant-slope Spring shows the opposite association to its patch responses. Holmes orientation under that cue varies with little monotone association. Censoring and native-field flags restrict interpretation even where rank correlation is strong.

Sources and reproducibility

Current figure legend and gate definitions

Supplementary material and supporting analyses

Earlier supporting material: S2. Numerical qualification and preparation provenance

Exact common starting states, duration-boundary audits, corrected deBelly endpoint controls and selected mesh diagnostics; source histories remain distinguishable from compact committed summaries.

polarity1d/manuscript/THREE_TESTS.md

Earlier supporting material: S3. Mechanism and stochastic sensitivity

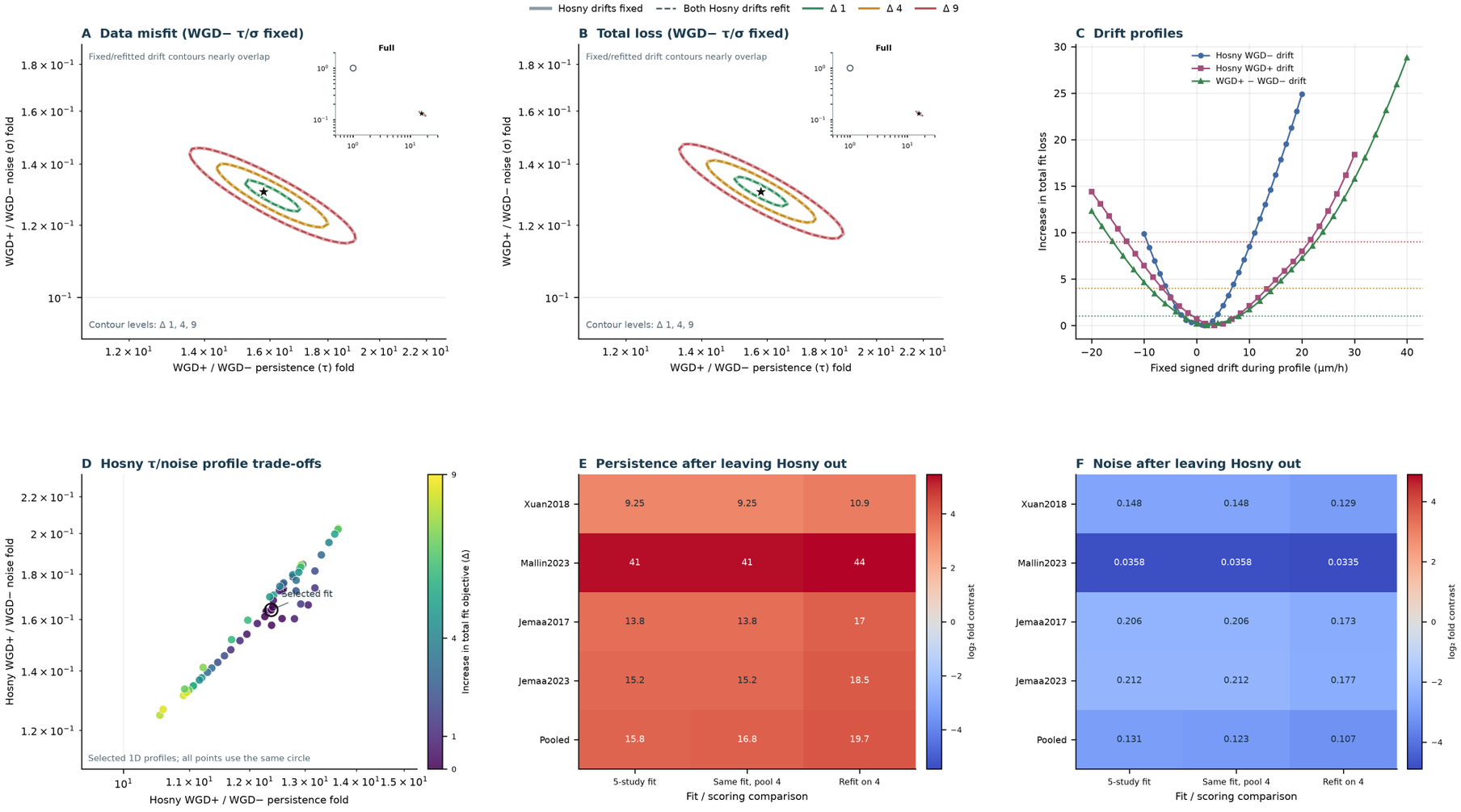
Complete gain/recovery grids, clock qualification, finite-cue definitions, outcome tables, site/vector diagnostics and numerical/noise controls accompany the mechanism checkpoint. The historical Figure 4 edition also displays both strengths, complete half-window diagnostics, physical-site eligibility and matched C20/C60 native examples; compact source tables support portable rerendering.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/VISION_CHECKPOINT_REQUIREMENTS.md

S4. Migration-model adequacy and Hosny sensitivity

How much does the Hosny drift assumption change the fitted contrast?

Primary Gaussian OU hierarchy · profile panels reoptimize remaining parameters · Δ values are descriptive fit sensitivity



Objective is arbitrarily weighted and has no calibrated statistical or biological units; it is not a likelihood. Profile ranges are unions of sampled 1D fits, not joint uncertainty regions. Hosny contributes medians to fitting.

Existing Hosny sensitivity supplement from the refined OU analysis.

Earlier supporting material: S4. Migration-model adequacy and Hosny sensitivity

Supporting conditional sensitivity analysis; it is not an independent biological replication.

Detailed legend

The retained supplement explores signed-drift and leave-Hosny-out sensitivity within the declared OU hierarchy. Its fitted changes and descriptive-score profiles are conditional on the model and supplied summaries, not confidence intervals. Data-only identifiability, Xuan population-curve residuals and prior/pooling assumptions remain part of the adequacy assessment. Representative movie diagnostics are separate: acquisition timing is uncertain and nuclear appearance does not establish ploidy.

Sources and reproducibility

Original Hosny supplement caption

Sensitivity methods and results

Earlier supporting material: Gain and small-loss recovery: supporting evidence

Detailed legend

Calculated native input gain is separate from observed post-cue activity. The small-loss recovery contrast persists through stored finite trajectories but weakens with time; actual damage is only about0.22–0.285% of core activity, with no reporter calibration.

Scope, limitations and pending validation

Small-loss assays are not the finite optical challenge or a universal stability measure.

Sources and reproducibility

Finite-time and amplitude audit

All97 gain/recovery values

Earlier supporting material: Selected native pulse and causal controls

Detailed legend

WP L15 T20/T60 starts from the same basal state and illustrates withdrawal-dependent domain survival. Evolved interventions have endpoint- and cue-specific limits.

Scope, limitations and pending validation

WP4× mobility passes orientation only at one dose; both conditions fail the stronger full-polarity-strength gate. Mobility is changed during cue and recovery.

Sources and reproducibility

Retained WP short/long pulse artwork

Endpoint-qualified controls

Supplementary material and supporting analyses

Earlier supporting material: Stationary-cue endpoint-vector alternative

An alternative view displays all 240 measured endpoint vectors on equal coordinate scales, with weak-resultant, overlap and finite-tail flags retained. It is available for discussion without replacing the complete withdrawal/retention map in the current review.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/standard_angles_sizes/vector_grid/README.md

Earlier supporting material: Spring: every native medoid

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_all_histories.png

Earlier supporting material: Spring magnitude, geometry and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_magnitude_geometry_controls.png

Earlier supporting material: Spring full/half-window memory

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_memory_windows.png

Earlier supporting material: Spring increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_increment_tails.png

Earlier supporting material: Spring short/long-lag geometry and circular mixing

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_endpoint_lag_sensitivity.png

Earlier supporting material: Spring largest sampled increment native context

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_large_increment_context.png

Earlier supporting material: All chemical models including Goryachev

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_stochastic_revision.png

Earlier supporting material: Chemical memory: both strengths

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_both_strengths_memory.png

Supplementary material and supporting analyses · continued

Earlier supporting material: Chemical magnitude and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_magnitude_controls.png

Earlier supporting material: Chemical increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_increment_tails.png

Earlier supporting material: Chemical site-gate sensitivity

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_site_gate_sensitivity.png

Earlier supporting material: Original chemical medoids, all18 conditions

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_selected_histories.png

Earlier supporting material: Goryachev lowest-resultant diagnostic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_lowest_resultant_diagnostic.png

Earlier supporting material: Spring active amount and spatial first harmonic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_amount_spatial_balance.png

Earlier supporting material: Preserved complete mechanism and assay checkpoint

The preceding outline retains all-five mechanism-to-measurement sheets, ring gain/recovery and finite-cue maps, and noise magnitude, controls and full/first/last-window diagnostics. Its numerical evidence is preserved; the revised main-figure claims are stated in this edition.

manuscript_outline/outline_v2.pdf

Prior all-five mechanism evidence crosswalk

Prior complete ring and noise checkpoint

Prior physical motion, magnitude and coverage evidence

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification
Figure 2 October 20 revision brief
Figure 3 October 20 revision brief
Figure 4 October 20 revision brief